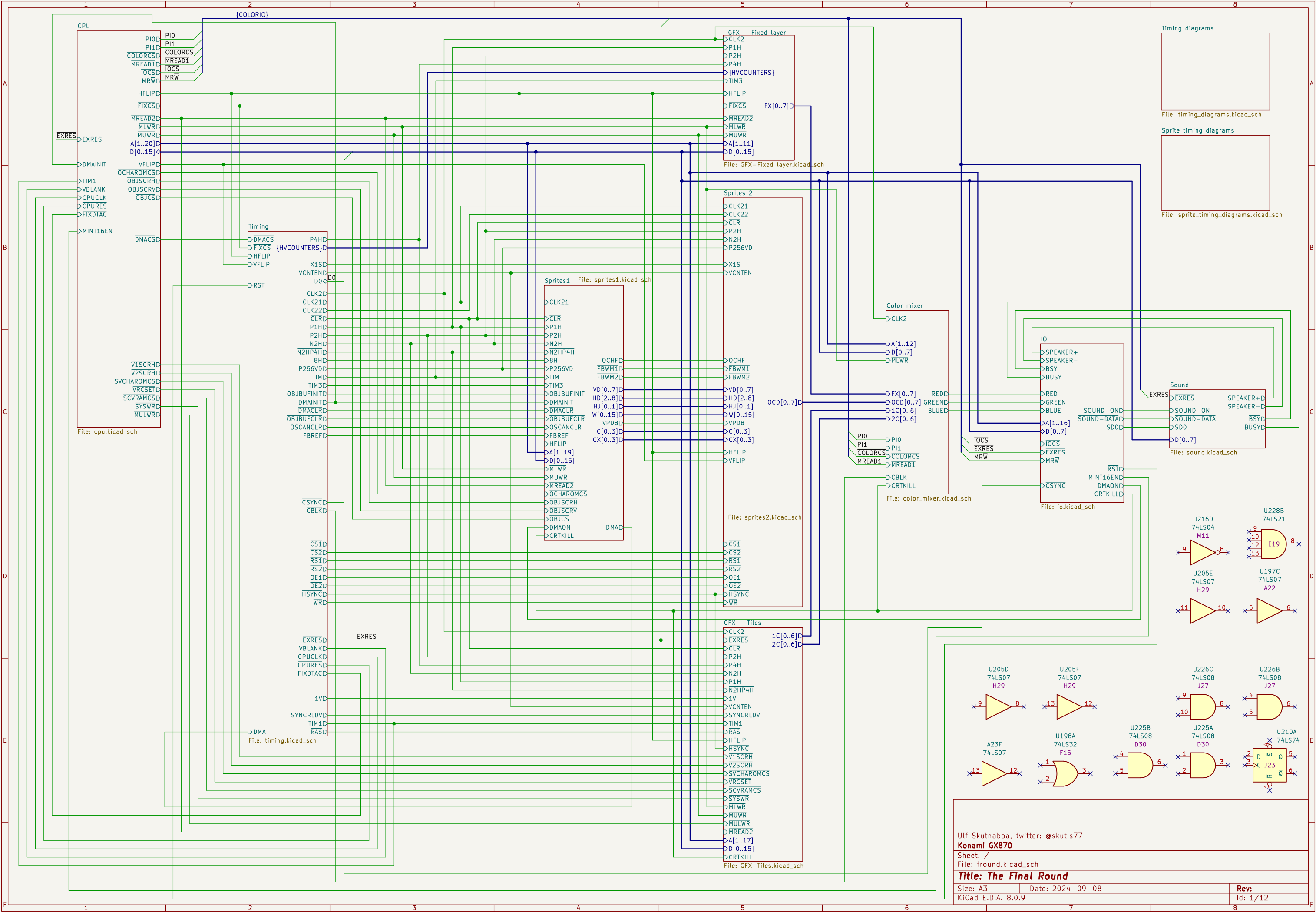
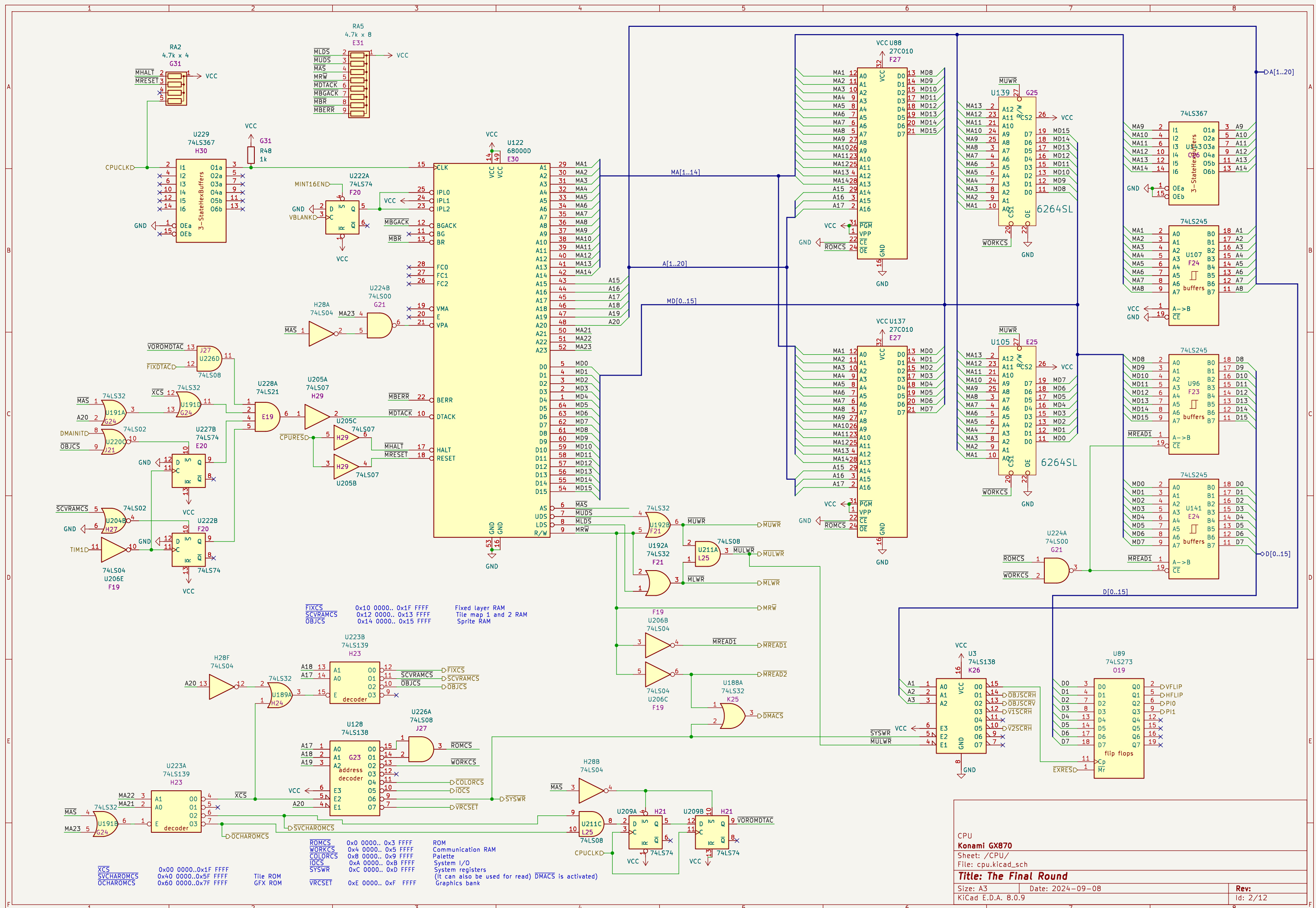
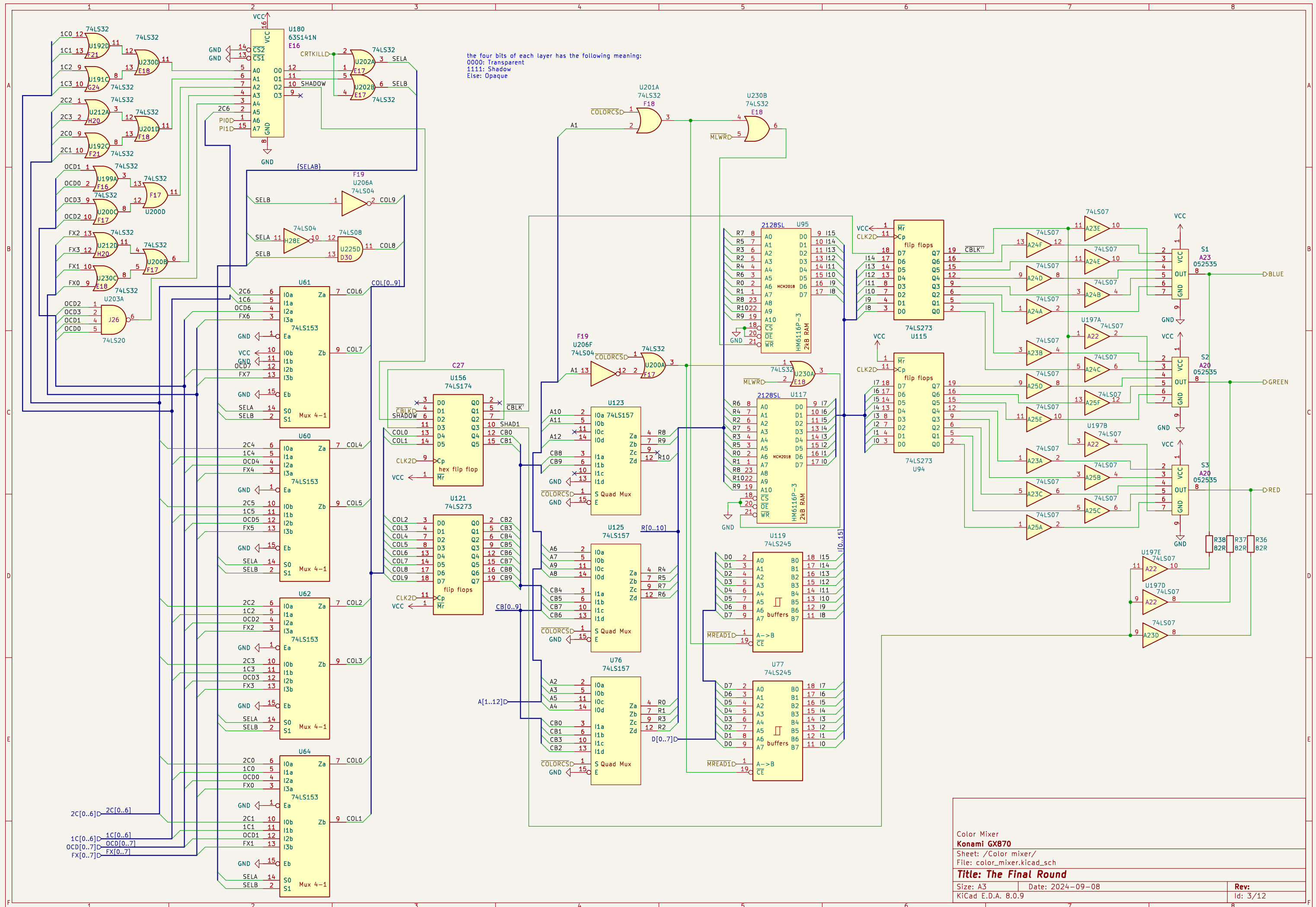
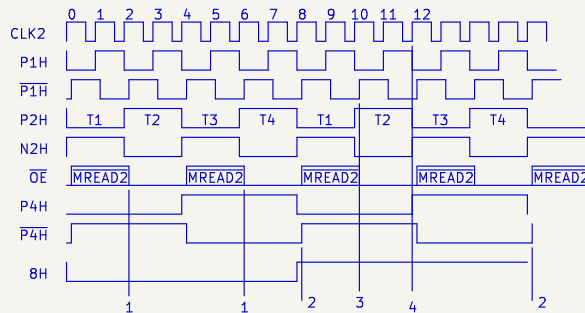








CLK2: 6.1397MHz
1V: 8.00kHz
2V: 4.00kHz
4V: 2.00kHz
P1H: 3.0719MHz
P2H: 1.536MHz
P4H: 0.767971MHz
8H: 384.000kHz
16H: 192.000kHz



RAM Control

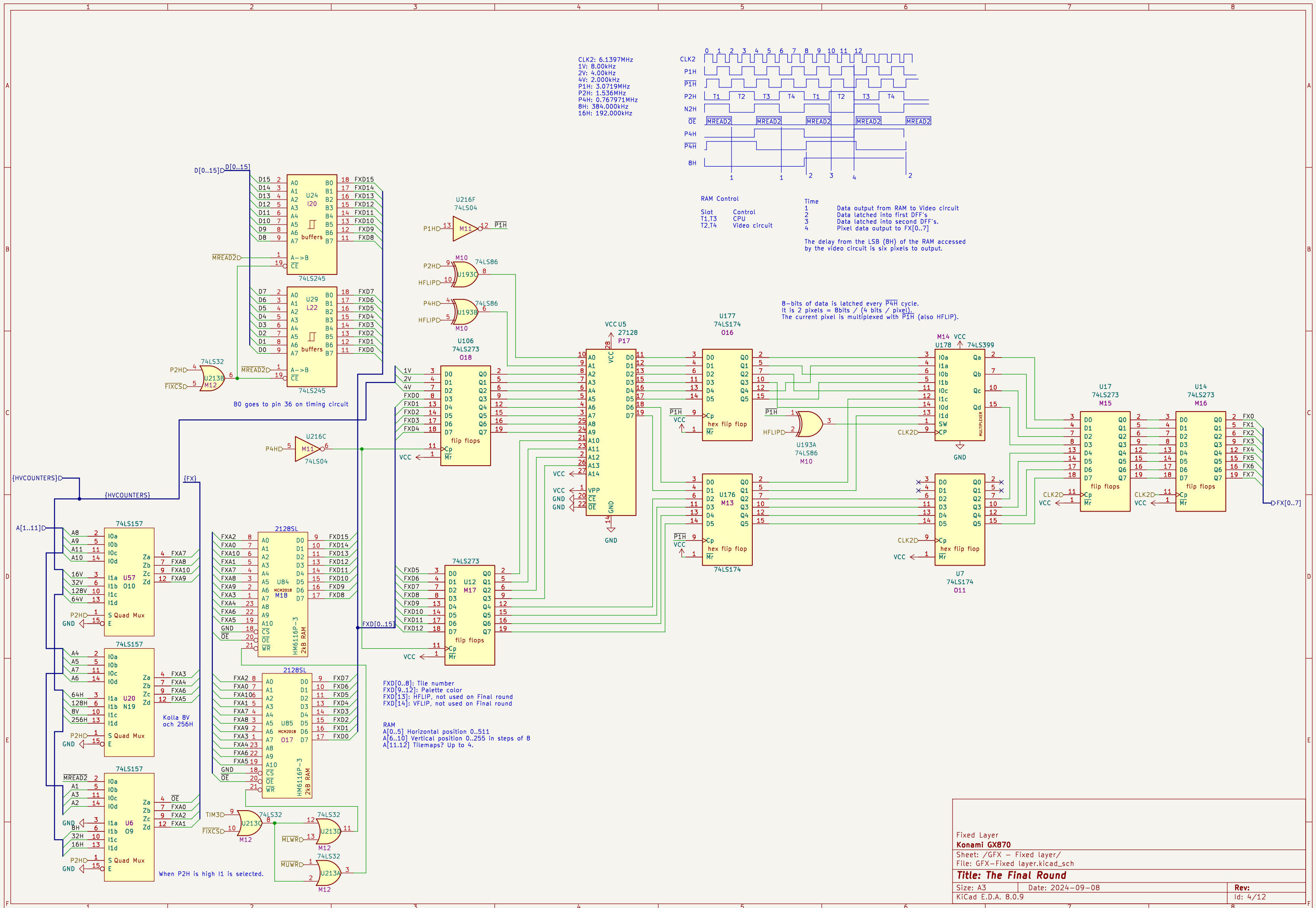
Slot	Control
T1,T3	CPU
T2,T4	Video circuit

Time

Time	Control
1	Data output from RAM to Video circuit
2	Data latched into first DFF's
3	Data latched into second DFF's.
4	Pixel data output to FX[0..7]

The delay from the LSB (8H) of the RAM accessed by the video circuit is six pixels to output.

8-bits of data is latched every P4H cycle.
It is 2 pixels = 8bits / (4 bits / pixel).
The current pixel is multiplexed with P1H (also HFLIP).



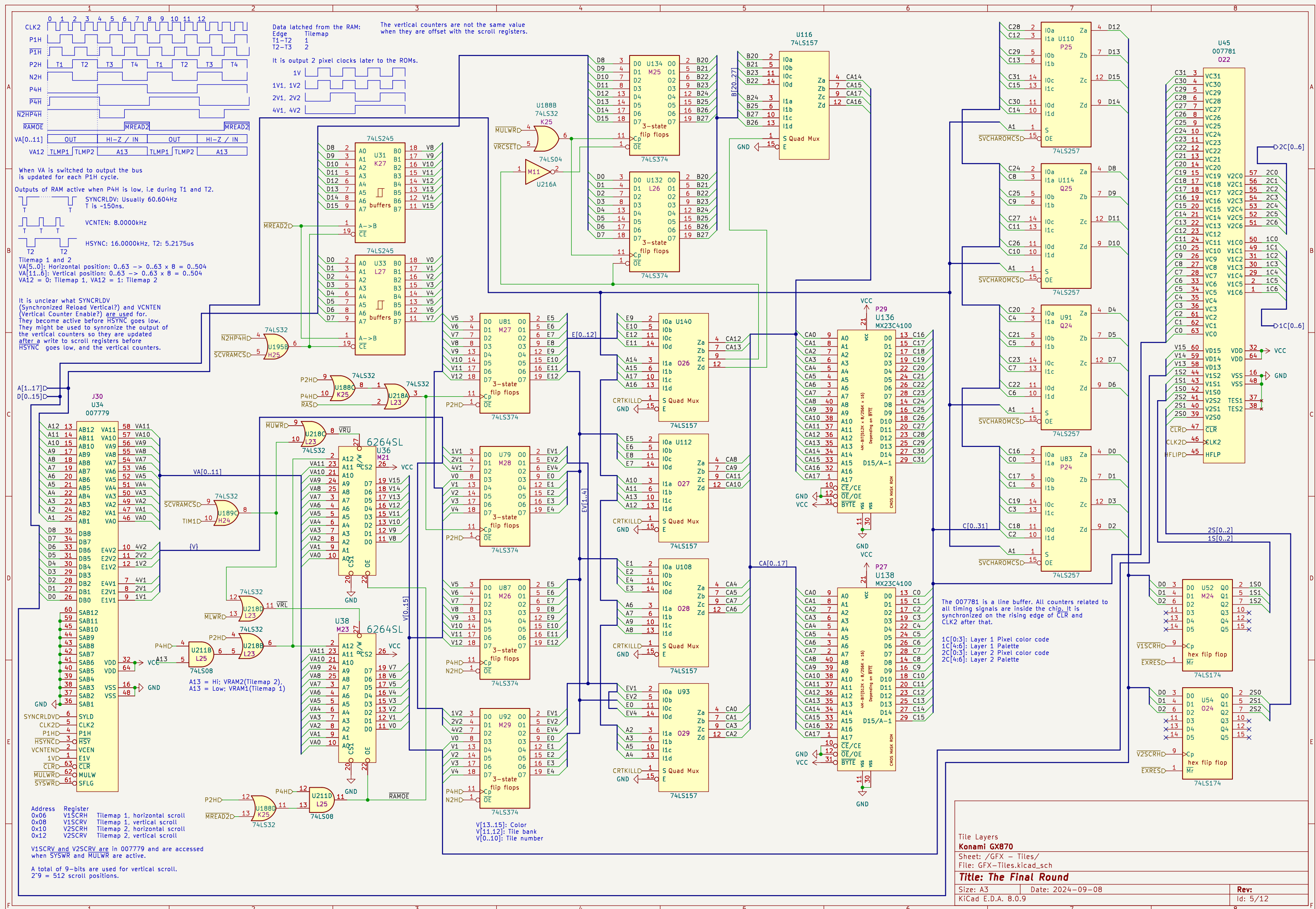
Fixed Layer
Konami GX870

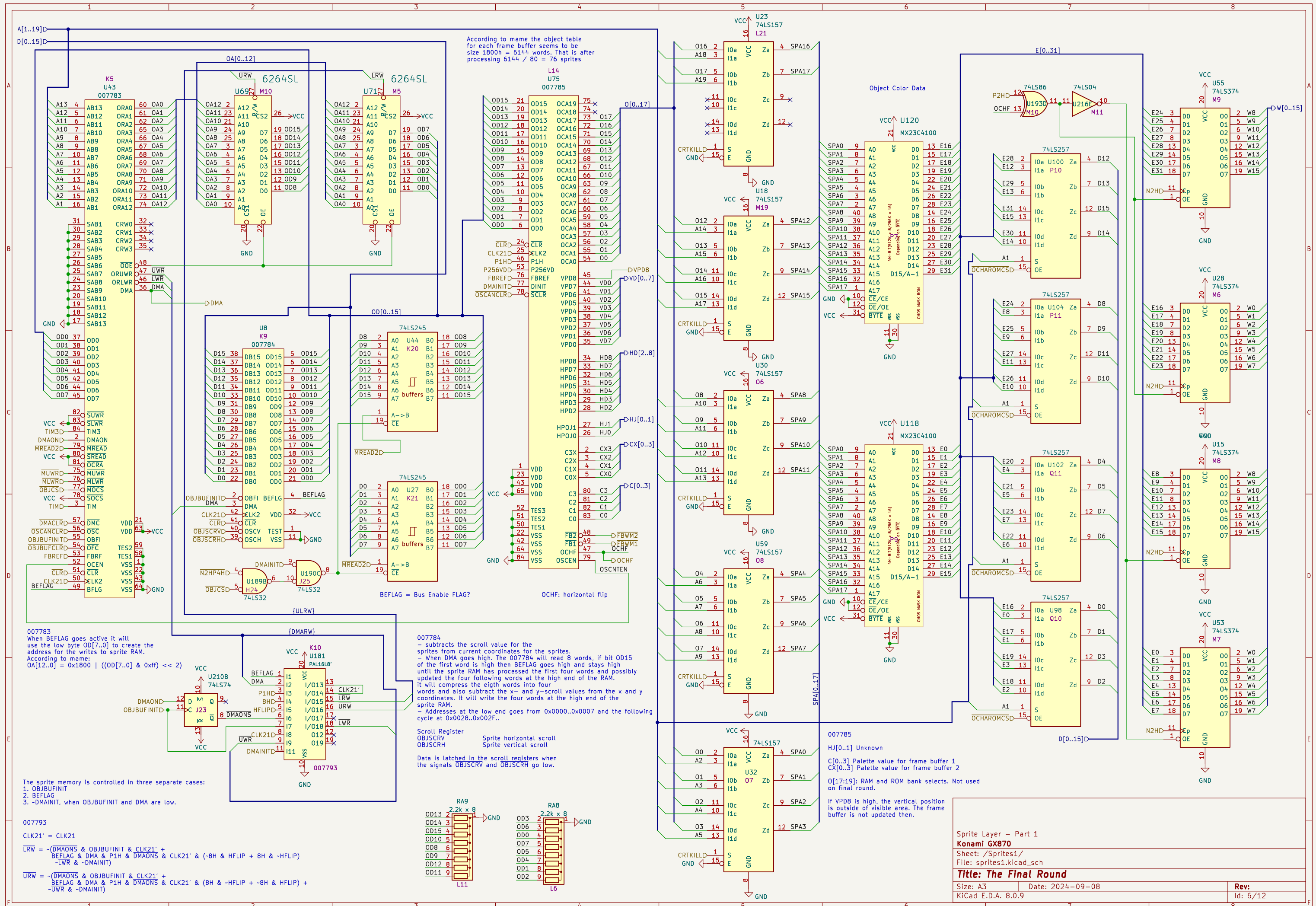
Sheet: /GFX - Fixed layer/
File: GFX-Fixed layer.kicad_sch

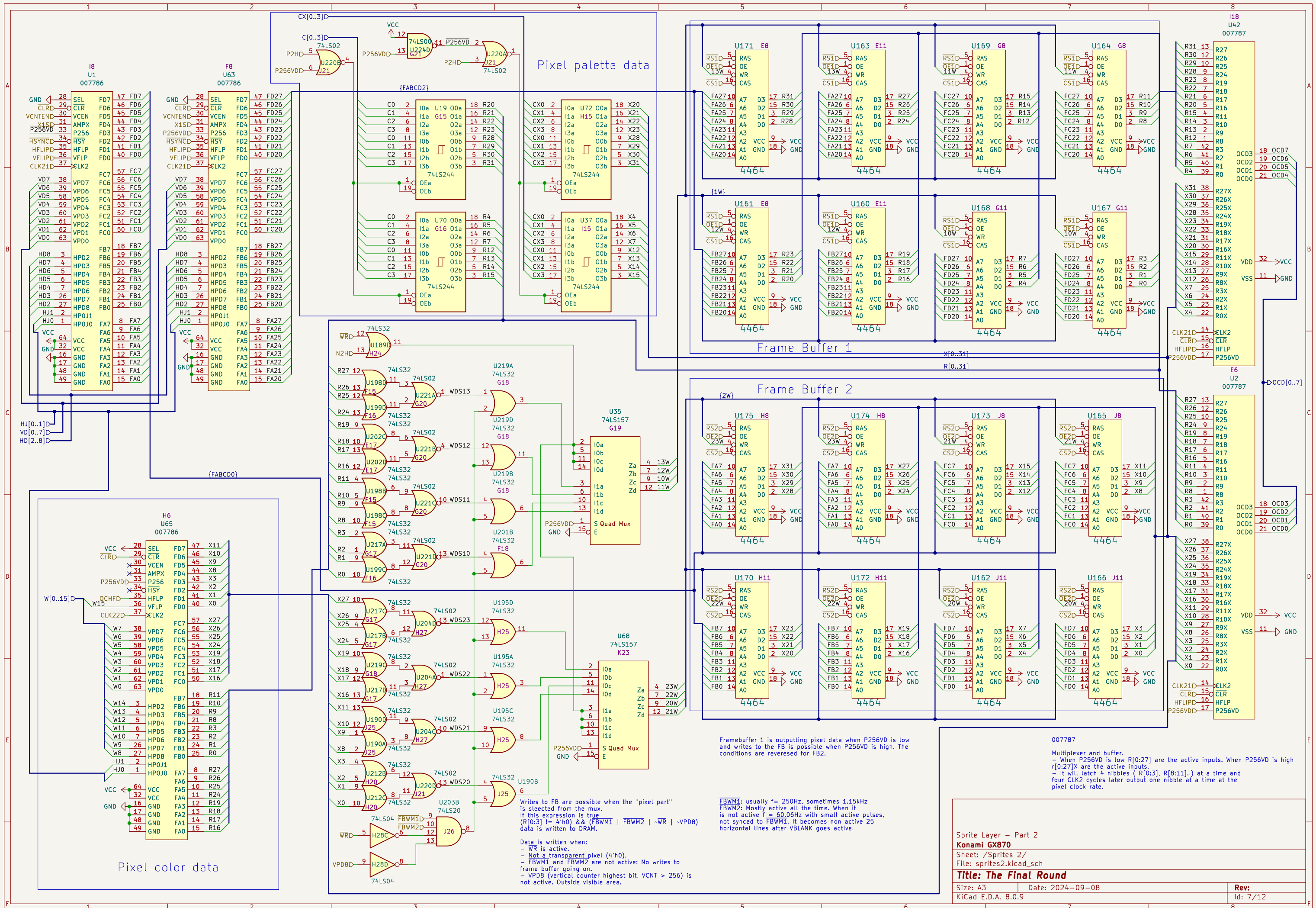
Title: The Final Round

Size: A3
Date: 2024-09-08
KiCad E.D.A. 8.0.9

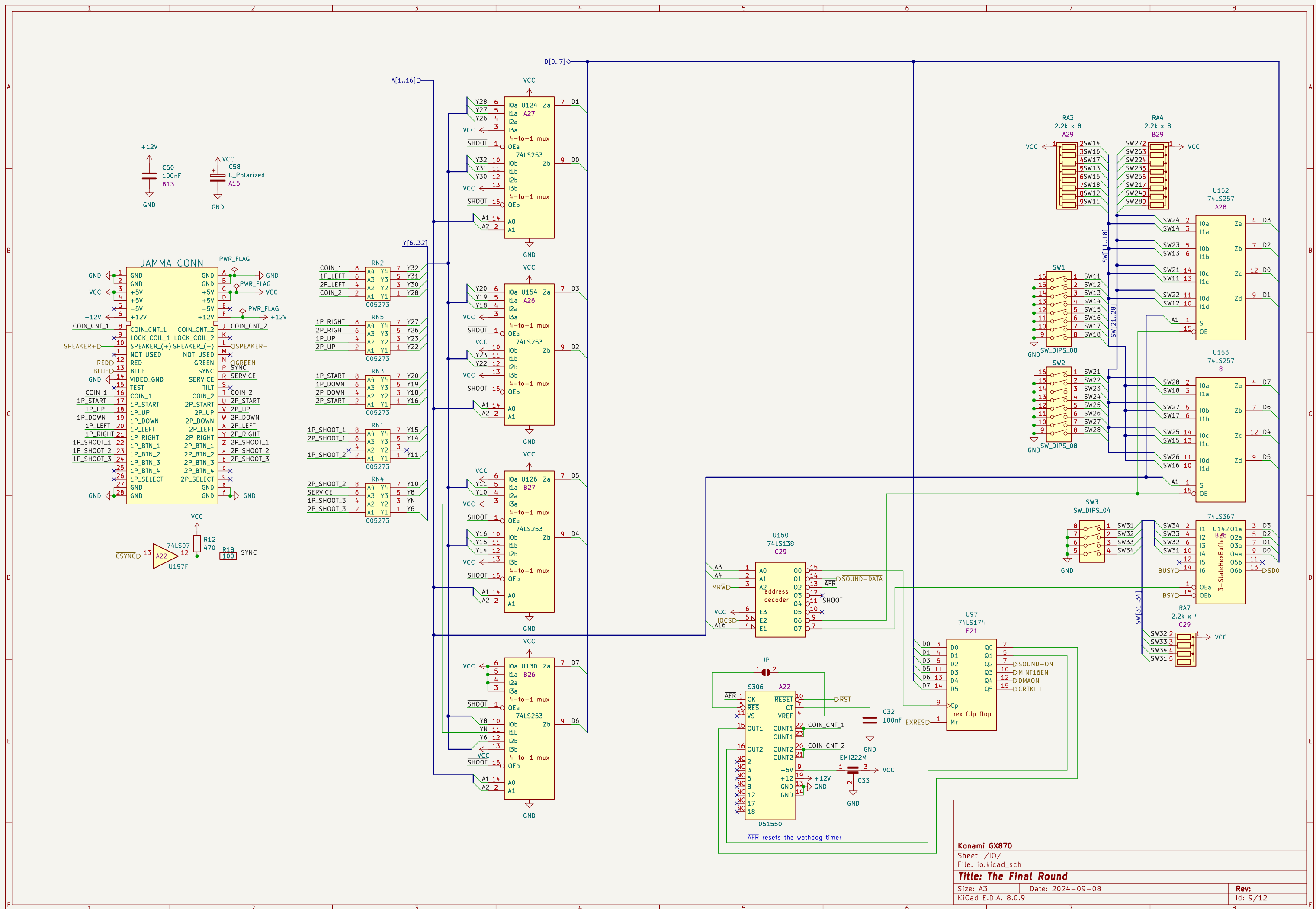
Rev:
Id: 4/12

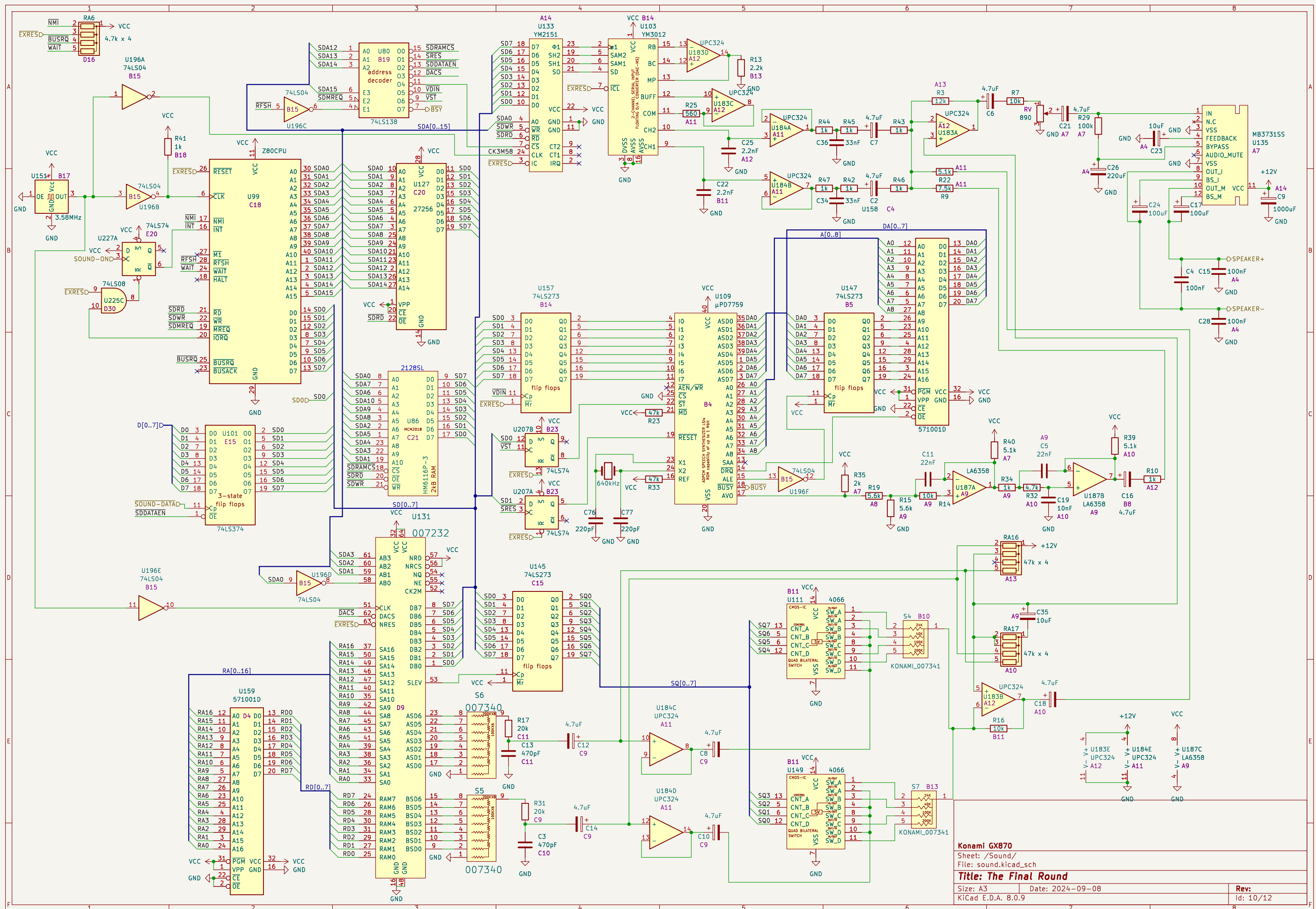












Horizontal signals

Frequencies:

- $f = \text{CLK2} / 8 = 768\text{kHz}$
- $f = \text{CLK2} / 16 = 384\text{kHz}$
- $f = \text{CLK2} / 32 = 192\text{kHz}$
- $f = \text{CLK2} / 64 = 96\text{kHz}$
- $f = \text{CLK2} / 128 = 48\text{kHz}$
- $f = 2/3 \times \text{CLK2} / 128 = 32\text{kHz}$
- $f = 2/3 \times \text{CLK2} / 256 = 16\text{kHz}$
- $f = 2/3 \times \text{CLK2} / 256 = 16\text{kHz}$
- $f = 16\text{kHz} / 2 = 8\text{kHz}$
- $f = 16\text{kHz} / 2 = 8\text{kHz}$
- $f = 16\text{kHz} / 2 = 8\text{kHz}$

Signal Legend:

- CLR
- P4H
- 8H
- 16H
- 32H
- 64H
- 128H
- 256H
- HSYNC
- HBLK ($\overline{\text{CBLK}}$)
- CSYNC
- VCNTEN
- 1V

The first HSYNC is at HCOUNT 175 after reset.

Notes:

- The first VCNTEN is skipped after reset. It goes low 140ns before HSYNC goes low, and high again when HSYNC goes low. VCNTEN is active right before every second falling edge of HSYNC.
- $\overline{\text{CPURES}}$ goes high, and stays high, on the seventh falling edge of HSYNC.

- $\overline{\text{CPURES}}$ goes high, and stays high, on the seventh falling edge of $\overline{\text{HSYNC}}$.

Vertical signals

$f = 16\text{kHz} / 2 = 8\text{kHz}$ $f = 16\text{kHz} / 4 = 4\text{kHz}$ $f = 16\text{kHz} / 8 = 2\text{kHz}$	1V 2V 4V
--	---

The diagram illustrates the timing of various video signals. The signals shown are:

- CTR**: Clock signal, 16kHz.
- f = 16kHz / 4 = 4kHz**: 2V signal.
- f = 16kHz / 8 = 2kHz**: 4V signal.
- f = 16kHz / 16 = 1kHz**: 8V signal.
- f = 16kHz / 32 = 500Hz**: 16V signal.
- f = 16kHz / 64 = 250Hz**: 32V signal.
- f = 16kHz / 128 = 125Hz**: 64V signal.
- f = 16kHz / 256 = 62.5Hz**: 128V signal.
- VSYNC**: Vertical Sync signal.
- VBLANK**: Vertical Blank signal.
- FBREF**: Frame Buffer Reference signal.
- SYNCRLDV**: Sync Reference Low/Valid signal.
- OBJBUFINIT**: Object Buffer Init signal.
- OBJBUFCLR**: Object Buffer Clear signal.
- OSCANCLR**: Oscillator Clear signal.
- P256VD**: 256V signal.

The diagram is divided into two sections: 'C' (Control) and 'D' (Data). The 'C' section shows the timing of the control signals, and the 'D' section shows the timing of the data signals. The diagram includes annotations for the period of the signals and the relationship between the signals.

Key annotations:

- Every 16th pulse is long.
- Every 8th pulse is long.
- Every 4th pulse is long.
- Every 2nd pulse is long.
- High period is longer
- VBLANK High: $7 \times 16 \times 2 = 224$, Low: $5 \times 4 \times 2 = 40$
- SYNCRLDV goes low 140ns before FBREF goes low. SYNCRLDV goes high again when FBREF goes low.
- OBJBUFCLR goes low one pixel clock before the rising edge of VBLANK and high again when VBLANK goes high.
- 60.606Hz, OSCANCLR goes low for 140ns - One pixel clock cycle.
- P256VD goes high/low one P4H cycle (8 pixels) after VSYNC goes high.

FBREF is measured at pin 22. At U186:3 FBREF is delayed by 22ns going high and 12ns going low.

The numbers in the HSYNC and HBLK diagrams are HSYNC cycles.
All edges are synchronised to the rising edge of CLK2.

Pixel clock CLK2 = 18.432MHz / 3 = 6.144MHz

Horizontal lines
HSYNC freq = 6.144MHz / 384 = 16kHz
HSYNC period = 1 / 16kHz = 62.5us
Pixels / line: 384
Active pixels: 320
Blank pixels: 64

Vertical lines
VSYNC freq = 16kHz / 264 = 60.606060Hz
VSYNC period = 1 / f = 264 / 16kHz = 16.5ms
scanlines / frame: 264
Active lines: 224
Blank lines: 40

The numbers in the VSYNC and VBLK diagrams are HSYNC cycles.
All edges are synchronised to the falling edge of HSYNC.

HCOUNT is bits [256H, 128H, 64H, 32H, 16H, 8H, P4H, P2H, P1H]
VCOUNT is bits [128V, 64V, 32V, 16V, 8V, 4V, 2V, 1V]

CBLK^{??} is at the output of color mixer.

Front Porch (FP) 24 px.
Back Porch (BP) 8 px

Front Porch (FP) and
Back Porch (BP) 16 HSYNC cycles.

Figure 10. HSYNC and VBLK Timing

The diagram illustrates the timing relationship between HSYNC and VBLK signals. The HSYNC signal (blue) has a period of 352 cycles, with a back porch (BP) of 8 cycles and a front porch (FP) of 24 cycles. The VBLK signal (red) has a period of 224 cycles, with a back porch (BP) of 16 cycles and a front porch (FP) of 8 cycles. The HCOUNT and VCOUNT signals are shown as counters. HCOUNT is a 16-bit counter from 333 to 495, and VCOUNT is a 16-bit counter from 239 to 15. The diagram is labeled 'Figure 10. HSYNC and VBLK Timing'.

Front Porch (FP) and Back Porch (BP) 16 HSYNC cycles.

HCOUNT is bits [256H, 128H, 64H, 32H, 16H, 8H, P4H, P2H, P1H]
VCOUNT is bits [128V, 64V, 32V, 16V, 8V, 4V, 2V, 1V]

$\overline{\text{CBLK}}''$ is at the output of color mixer.

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